

Alternating Current

13.3

You know that Faraday's law of electromagnetic induction requires a changing magnetic field to produce an electric current. If you push a permanent magnet into a coil, an electric current is produced while the magnet is moving. Lenz's law predicts the direction of the current. However, the current goes in one direction for only as long as you move the magnet in one direction. It is not possible for you to move the magnet into the coil in the same direction indefinitely. At some point, you will need to pull the magnet in the opposite direction. As soon as you reverse the direction of the magnet, the electric current also reverses direction. A current that periodically reverses direction is called an **alternating current**.

alternating current an electric current that periodically reverses direction

Development of Alternating Current

Recall from Section 11.5 that direct current is a flow of electrons in one direction only. To cause an electric current, a potential difference is applied across the circuit by a source of electrical energy, such as a battery. This causes the electrons present throughout the circuit to move in one direction. Charges flow from one terminal of the battery through the circuit and eventually back to the other terminal. This situation describes the current in the small circuits you studied in Chapter 11. Do the principles involved in these small circuits apply to a large circuit like the circuits in your home or even the electrical power grid?

Today's electrical power grid does not rely on direct current. The transfer of electrical energy using direct current is limited to how far it can be transferred without significant energy loss in the form of thermal energy. However, direct current was the standard used in the first electrical power grids. In 1882, Thomas Edison built the Pearl Street power station in Manhattan to illuminate homes and stores. He set up an electrical power grid using direct current, but he was only effectively able to transfer electrical energy to 193 buildings.

Nikola Tesla, an inventor and electrical engineer, developed a competing system that used alternating current. The alternating current system could transfer energy from a power plant more efficiently than the direct current system could. In 1896, the Edward Dean Adams Station, a United States hydroelectric power station, delivered the first alternating current along a power grid to the city of Buffalo.

Edison and Tesla both fought to have their systems accepted, but in the end alternating current became the favoured choice. In 1922, the Sir Adam Beck Generating Station started producing alternating current electricity in Niagara Falls, Canada. 

WEB LINK

To find out more about Tesla's and Edison's battle for electrical power distribution,



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What Is Alternating Current?

Alternating current is a back and forth motion of charges. To understand alternating current, look at the graphs of current and voltage versus time in **Figure 1**.

As the voltage, V , increases, the conventional current, I , in the wire increases in a positive direction. The voltage increases until it reaches a maximum positive value. At the same instant, the current also reaches a maximum positive value. The voltage then decreases through zero until it reaches a maximum negative value. The current similarly decreases.

In Figure 1, the process repeats at a frequency of 60 Hz, which is the frequency of electricity in the North American electrical power grid. This means that the current is going in a positive direction, reversing, and going in a negative direction 60 times each second. So the electrons in the wire never travel the complete length of the circuit; rather they move back and forth in about the same spot. Note that the graphs both go through zero twice during each cycle as the current is changing directions. This implies that the circuit is effectively off for an instant. You may ask why we do not notice this effect. In fact, lights do get dimmer and brighter during each cycle, but the cycles occur so quickly that our eyes cannot detect them.

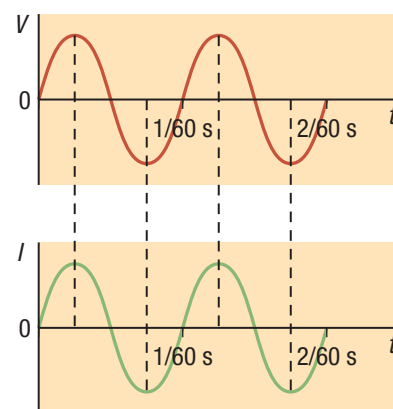


Figure 1 Voltage and current versus time for an alternating current. Note the frequency of 60 Hz.

Since the electrons move back and forth in about the same spot, you may think the current is ineffective at transferring energy. To understand how an alternating current can cause an energy transfer, consider a clothes washing machine. In a washing machine, an agitator swishes the wet clothes back and forth. The agitator still causes changes in energy, and thus work is done to clean the clothes. In an alternating current circuit, electrons move back and forth to cause energy changes in the electrical device, and thus work is done.

All the relationships that you have learned involving circuits and their components, for example, Ohm's law, still apply with alternating current. In the graphs in Figure 1 on the previous page, as the voltage increases, so does the electric current. This is in accordance with Ohm's law, and the current depends on the resistance of the wire that is used to carry the alternating current.

Household Circuits

There are limits to how much alternating current homes can draw. Smaller homes and apartments may only require 50 A to 100 A of electric current. Larger homes require 150 A to 200 A of electric current. The wire that enters your home from the electrical power grid is called the service line, and the amount of current it carries is labelled at the point of entry. For example, you may have a 100 A service going to your home. The service enters your distribution panel, which controls all the circuits in your home (Figure 2).

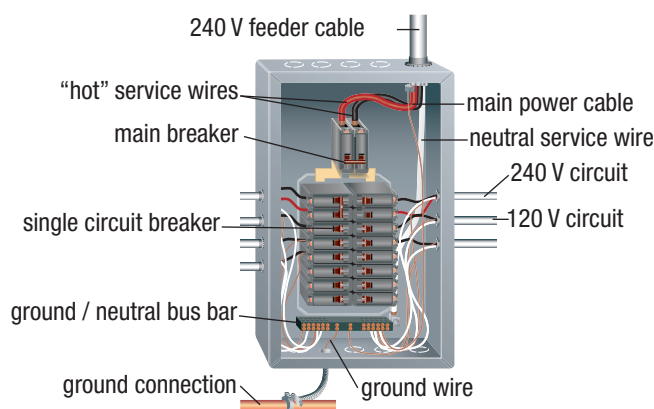


Figure 2 A distribution panel controls all of the circuits in your home. You can think of this panel as the power supply for your home. Note the red, black, and white service lines.

The AC voltage oscillates between a maximum positive voltage and a maximum negative voltage. Since the maximum voltage is reached for a brief instant twice during a cycle, we do not state that voltage. Instead, we use a value that is called the root mean squared or RMS value. This is an effective voltage that is available most of the time during a cycle. It also allows for the use of one voltage in calculations instead of a constantly changing voltage.

Your home is designed so that an effective voltage of 240 V is applied to the electrons in the wires in your home. Only a few of your household appliances require 240 V, such as an electric stove or clothes dryer. Most appliances require an effective voltage of only 120 V. To accommodate both requirements, a three-wire system is used. Two of the wires are considered “hot” and are coloured red and black. The third wire is white and is called the neutral wire. The voltage between the black and the white, and the red and the white, is 120 V. The voltage between the red and the black is 240 V. This way, an electrician can wire the home's circuit with just two of the wires. Typically, the electrician uses only the black and white wires when 120 V is required. If an appliance requires 240 V, then the black and red wires are used.

CAREER LINK

Electricians must have an understanding of the electrical requirements of appliances to wire homes. To find out more about becoming an electrician,



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A fourth wire, called the ground wire, is another essential part of the wiring. Ground wires can be bare (not insulated) or green and are electrically connected to the ground via your electrical distribution panel. The purpose of the ground wire is to prevent stray currents from reaching you when you touch a circuit and to direct them safely into the ground. As a safety feature, all electrical receptacles and switches must be grounded.

The receptacles for 240 V appliances and 120 V appliances are different. A 240 V plug may have four prongs, one for each of the wires: red, black, white, and bare (or green) (**Figure 3**). Most 120 V plugs require three prongs, one each for black, white, and ground. There are also many devices, such as lamps, that use two-prong plugs and do not have a ground pin.

Some electronic devices (such as laptop computers) require direct current because they cannot operate with alternating current. These electronic devices need an adapter that converts the alternating current into direct current. The electrical requirements of some common household appliances are listed in **Table 1**.



Figure 3 A plug for an electric stove or clothes dryer is designed for a maximum alternating current of 30 A at 240 V. Since it has four prongs, it can also supply 120 V for parts of the appliance that do not require 240 V (such as control circuits).

Table 1 Electrical Requirements of Some Common Household Appliances

Device	Type of current required	Current requirements (A)	Voltage requirements (V)
electric stove	AC	30	240
clothes dryer	AC	30	240
air conditioner	AC	20	240
iron	AC	5	120
hair dryer	AC	12.5	120
refrigerator	AC	6	120
40-inch LCD television	DC	18	12
desktop computer (excluding monitor)	DC	0.5 to 20 (depends on the component)	0.8 to 12 (depends on the component)
telephone charger	DC	1	5

Safety Systems

Many safety systems exist in your home to prevent harm to you, damage to your appliances, or electrical fires. These include fuses, circuit breakers, ground fault circuit interrupters, and arc fault circuit interrupters.

Fuses

Fuses are devices that are placed in series with one of the parallel branches in your home. If the current exceeds the maximum value at which the fuse is rated, the wire inside the fuse melts, which opens the circuit and prevents any more current. Fuses are for single use only. Fuses may also be found inside appliances as a further measure of protection.

Circuit Breakers

Circuit breakers are devices that prevent too much current in a wire for an extended period of time. If too much current is in a wire, the wire will heat up and the insulation may melt and potentially cause a fire. Circuit breakers work by using a strip of two different metals fused together. The strip of fused metals is called a bimetallic strip.

When too much current is present, the bimetallic strip heats up and bends. This causes the breaker to “trip,” which turns off the circuit. If a circuit breaker trips, you should consider why it tripped. It could be because too many loads are connected in parallel to the same circuit and require too much current. Using fewer electrical devices on that circuit should prevent the tripping. Circuit breakers are reusable and can be reset.

Ground Fault Circuit Interrupters

Ground Fault Circuit Interrupters (GFCIs) are installed in places like bathrooms. Their purpose is to detect any differences in the current going into a circuit compared to the current going out. If the GFCI does detect a difference, it immediately shuts off and stops the current. Suppose that you touch an electrical outlet with a wet hand and then reach for an appliance with another wet hand. It is possible that you may set up a circuit pathway for electric current to travel through your body. This could cause electrocution. The GFCI very quickly detects the difference in current and immediately turns off the circuit. GFCIs respond much more quickly than regular circuit breakers because they are designed to trip with a small current fault, whereas regular circuit breakers will only trip if the current exceeds the maximum rating. For example, a GFCI will trip if a current difference of 0.006 A is detected, whereas a current of over 15 A needs to be detected to trip a regular breaker.

Arc Fault Circuit Interrupters

Arc Fault Circuit Interrupters (AFCIs) prevent sparking or arcing, which could start a fire. Arcing occurs when electric current travels through the air and causes a spark. This may happen if the insulation around the wiring has become frayed. The bare wire can possibly move close to a metal part and cause arcing to occur. The temperature of an arc is very high and could cause a fire. Arcing may not be enough to cause a circuit breaker to trip, so the AFCI prevents current flow if an arc is detected. AFCIs also act more quickly than regular circuit breakers.

13.3 Summary

- Alternating current is an electric current that periodically reverses direction.
- Alternating current frequency is 60 Hz in the North American electrical power grid.
- Some appliances require alternating current, whereas others require direct current.
- Your home requires a certain amount of current at voltages of 120 V and 240 V.
- The circuits in your home are protected by fuses, circuit breakers, ground fault circuit interrupters, and arc fault circuit interrupters.

13.3 Questions

1. Do electrons travel from the power plant to your home to provide electrical energy? Explain your answer. K/U
2. In alternating current electricity, is the voltage proportional to the current? Explain. K/U
3. (a) Would you notice if the frequency of the alternating current electricity were reduced to 2 Hz? Explain.
(b) In the late 1950s, the frequency of alternating current was changed from 25 Hz to 60 Hz. Suggest a reason for this change. K/U C
4. How do 120 V plugs differ from 240 V plugs? K/U
5. A laptop computer requires 12 V direct current, and yet it is plugged into a home's wall outlet. What must be involved to satisfy the requirements of the laptop? K/U
6. Describe the differences between fuses, circuit breakers, GFCIs, and AFCIs. K/U
7. Most household circuits in North America use 120 V, while in Europe 240 V is most commonly used. Research to find out why this difference exists. Is one system better than the other? T/I



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